Methods and DEV measurements

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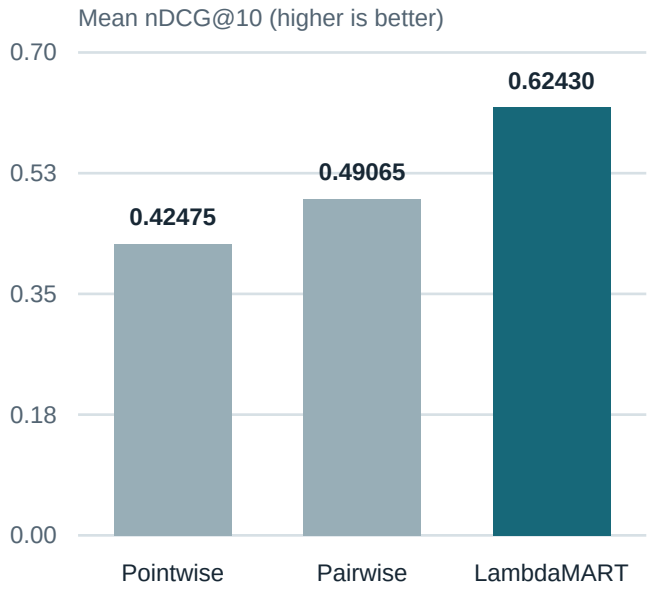
Ranking methods and data declarations

Fusion is handwritten, equal-weight min-max CombSUM over BM25 whole-document, BM25 title and LMIR Dirichlet whole-document. LMIR JM is disabled. The rule was selected on TRAIN only; RRF and CombMNZ retain their textbook definitions.

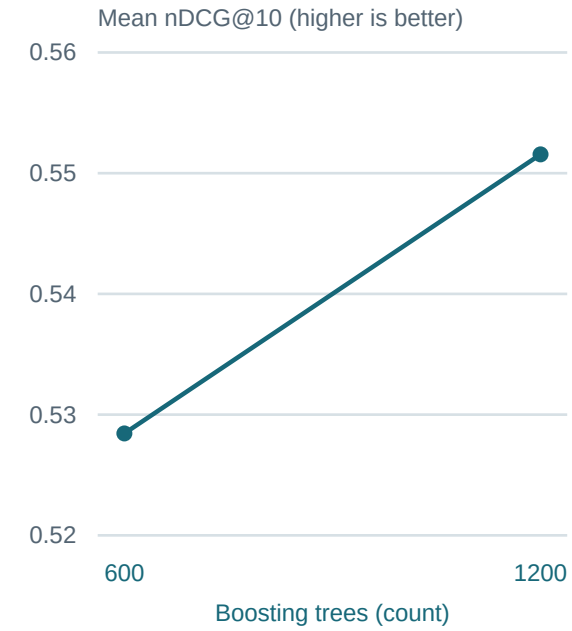
The main artifact is native LightGBM LambdaRank. Its declared feature schema is **raw_plus_query_percentiles**. It is fitted on 958,671 original TRAIN+DEV rows from 8,000 queries. Any augmented copies stay within complete query groups. The artifact loads without retraining.

Pointwise uses handwritten squared-error training with TRAIN gradients and DEV epoch selection. Pairwise is a refined plain 136-256-128-1 ReLU MLP within the staff width limits. It uses TRAIN+DEV gradients and frozen TRAIN-only standardization; no model-score distillation is used.

Loaded-artifact DEV accuracy



DEV versus tree count



Right plot: a single seed-0 augmented baseline, not the final blend. DEV supplied gradients; this curve measures fitted DEV accuracy.

Interpretation: DEV supplied gradients to the main and pairwise models, and stopping feedback to pointwise. These are development measurements, not independent generalization estimates. Loaded-artifact preflight also verifies finite scores, deterministic inference and the candidate/latency contract.

Fold1 TEST is excluded from the current fitting, augmentation, preprocessing and local selection. Earlier experiments consulted TEST in violation of the rules; refitting does not undo that history. The limitations are recorded accurately in docs/COMPLIANCE.md.

Grouped cross-validation

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Five query-disjoint folds use released TRAIN only. All documents for a query stay together. Query-relative features use no labels or corpus-fitted statistics. For augmented runs, augmentation occurs after the fold split. Each fit runs to the largest fixed tree count; reported folds are not used for early stopping.

Source: **top3_extra127_cv1200.json**. This is a 127-leaf baseline sweep, not CV of the exact final blend. Subsequent model changes also use reserved TRAIN development queries and declared DEV/portal feedback.

Fixed tree-count sweep: training versus unseen queries

Trees	Train mean	Validation mean +/- SD	Val / train
400	0.55366	0.47526 +/- 0.00777	0.8584
800	0.60332	0.47752 +/- 0.00794	0.7915
1200	0.63846	0.47774 +/- 0.00859	0.7483

The highest mean in this sweep is **0.47774** at 1200 trees. Its validation-to-training ratio is 0.7483. Growing training accuracy faster than validation accuracy exposes memorization of training queries. Fold standard deviations describe variation across these five query partitions, not a confidence guarantee.

The CV sweep and the reserved-query probes are development comparisons. Repeated selection, trained-on DEV measurements and portal feedback must not be represented as a fresh held-out test.

Configuration and reproducibility

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Final configuration and tradeoff

0.650 x 2000 trees, 127 leaves, seed 0, augmented, raw and percentile splits, randomized thresholds; 0.350 x 2400 trees, 63 leaves, seed 23, augmented, raw and percentile splits, randomized thresholds.

Leaf-model settings: learning rate 0.05, minimum leaf rows 100, L2 penalty 10, feature fraction 0.8, bagging fraction 0.8. Feature and row sampling with fixed seeds regularize the fits.

The configuration trades fitted DEV accuracy against generalization and robustness. Augmentation perturbs only TRAIN/DEV feature rows and retains relevance labels; query-relative transforms reduce reliance on absolute feature scales. Blending averages label-trained tree predictions within a native LightGBM artifact. More fitted DEV accuracy alone does not establish better accuracy on unseen queries.

Track C1 uses the generic recipe in submission/recipe.py: it reads only the two supplied benchmark input files, derives feature width from their schema, trains with a fixed seed and a training deadline, and writes a native model. Inference uses the artifact schema; it does not identify hidden datasets or branch on query IDs.

How to reproduce and verify

Install the grading requirements, prepare the released TRAIN/DEV files, then run the ordered reproduction commands in docs/CURRENT_MODEL.json. The main artifact SHA-256 is:

9f0d6bf8bfcedc76b70c309da327bd2d

daa1391e8a5430741179dffcd21cd49b

The neural settings and original TRAIN standardizer are documented in docs/CURRENT_NEURAL.json. Use the unchanged staff artifact loader to check the plain MLPs. Run the artifact regression checks and scripts/preflight.py on released DEV. The toy smoke test belongs in a disposable copy because it overwrites the main artifact.

Preflight: prepare 0.250 seconds; largest-query rerank 0.260 seconds. The measured largest-query call is below the 2-second cap; the staff grader remains authoritative for the whole-run limit.

The input audit matched cached features, labels, query IDs and recomputed transforms for all 6,000 TRAIN and 2,000 DEV queries. The combined cache matches TRAIN then DEV exactly. Source hashes and checks are in docs/TRAINING_AUDIT.json.

AI use and code provenance

Codex assisted with implementation, experiments, testing, documentation and authorized portal submissions; ChatGPT assisted with assignment analysis. Formula/code sources are the assignment and starter repository. LightGBM implements the permitted LambdaMART step; required fusion, pair construction and neural losses are handwritten. No external LTR implementation or teacher scores supervise the neural models. The student must understand and verify these declarations.